

Practice Midterm

Form P · Practice

Practice for the midterm exam · not graded · use it to time yourself

Name: _____

How this practice works

- **Same skills, different questions.** The real midterm follows the same six stages (think first, check each variable, check how variables relate, build models with LINEST, test on unseen days, advise) and is marked the same way. The questions and the results here are **different**: you will meet a variable whose sign may surprise you, two variables that cannot be separated, and an extreme day that the model cannot explain. If you can do every item here and explain why, you are ready. Memorising these answers will not help.
- **The scenario.** You are the analyst for a seaside café. The owner wants to forecast tomorrow's **pastry_units**.
- **Stage 1 comes first.** It asks what you expect **before** any data. Finish it before you turn the page.
- **Time and points.** 50 points, about 55 minutes. The suggested time for each stage is in its banner.
- **Show your work.** Partial credit is given for correct steps. Round to 2 decimal places unless told otherwise. A reference sheet is on the last page.
- **Allowed:** pencil or pen, eraser, ruler, calculator (a phone as a calculator only). **Not allowed:** notes, books, other use of a phone, talking.

Stage	What it covers	Points	Score
1	Think first: choose and defend your variables	9	
2	Check each variable on its own	7	
3	Check how the variables relate to each other	7	
4	Build and compare models with LINEST	13	
5	Does the model forecast unseen days?	7	
6	Advise the owner and extend the model	7	
Total		50	

About the data

The café recorded sales and weather on **168 consecutive trading days** (7 April–21 September 2025). The data are **synthetic teaching data**, simulated for this course, not real café records. The outcome to forecast is **pastry_units**: the number of standard pastries sold that day. Assume the café can get a reliable weather forecast and the beach-event calendar the evening before, and that it sets its own prices.

Table 1. Candidate variables you may use to forecast the outcome.

Variable	What it measures
temperature_c	Daily mean outdoor temperature (°C) during opening hours
sunshine_hours	Hours of sunshine during opening hours
rainfall_mm	Rain (mm) during opening hours; 0 means a dry day
wind_kmh	Mean wind speed (km/h) during opening hours
humidity_pct	Mean relative humidity (%) during opening hours
beach_event	1 if an organised public event is held on the nearby beach, 0 otherwise
weekend	1 if Saturday or Sunday, 0 otherwise (derive from day_of_week)
pastry_price_eur	Posted price (€) of one standard pastry. The café changed it once, on 30 June, and never tried other prices
post_change	0 before 30 June, 1 from 30 June (the day the café's new price policy began; derive from date)
gelato_units	Number of gelato servings sold that day
pastry_revenue_eur	pastry_price_eur × pastry_units (derive)

STAGE 1 · Think first: choose and defend your variables

≈ 8 min · 9 pts

Research question. Which variables from the table on page 1 should the café use to forecast tomorrow's **pastry_units**? Use only the table and how a café works: you have not seen any data yet.

- 1 Critique a colleague's model.** A colleague proposes the four variables below. For each, write **KEEP** or **DROP** and one reason based on how the café works (not on correlations).

4 pts

Variable	KEEP or DROP	Reason (a few words)
gelato_units		
pastry_revenue_eur		
beach_event		
humidity_pct		

- 2 Two hypotheses.** Write **two** hypotheses, each about a different variable you would use. Each must give the **direction** of the effect and be something data could show to be wrong. For one of them, describe a LINEST result that would show it was wrong.

3 pts

- 3 Validity or reliability?** Classify each scenario as a **validity** problem (the number does not capture the concept) or a **reliability** problem (the measurement would change if repeated or taken under different conditions). Then explain one choice in a sentence.

2 pts

Scenario	Validity or reliability?
(i) The weather sensor is moved from a shaded wall to a sunny wall in July, and April and August temperatures are compared directly.	
(ii) Two staff count the pastries left at closing and often disagree by a few.	
(iii) The owner treats "pastries sold" as a measure of how much customers enjoyed the pastries.	

Stop. Finish Stage 1 before you turn the page.

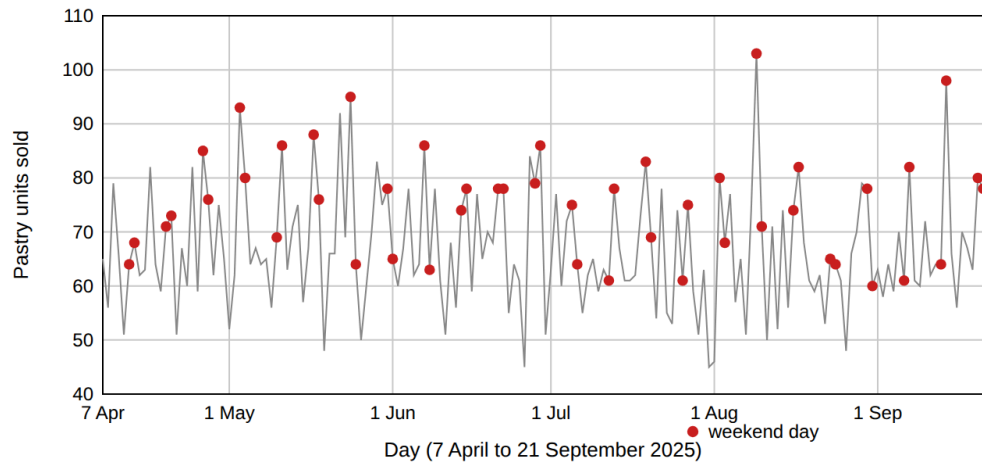
STAGE 2 · Check each variable on its own

≈ 7 min · 7 pts

Table 2. Summary of all 168 days (Sheets: AVERAGE, MEDIAN, STDEV.S, MIN, MAX).

Variable	Mean	Median	Std dev	Min	Max
temperature_c	22.80	22.90	3.43	12.6	31.8
sunshine_hours	7.17	7.35	2.48	1.0	13.0
rainfall_mm	0.76	0.00	1.92	0.00	14.00
wind_kmh	13.53	13.20	6.98	3.0	26.0
humidity_pct	66.58	67.00	8.56	45.0	95.0
pastry_price_eur	2.62	2.62	0.13	2.50	2.75
pastry_units	67.07	65.00	10.80	45.00	103.00
weekend (0/1)	0.29	0.00	0.45	0	1

Figure 1. Pastry units sold on each of the 168 days; weekend days are marked with red dots.



- 4

Read the plot. (a) What regular pattern do you see in Figure 1, and which variable in the table on page 1 would capture it? (b) The highest day is **103** pastries, on Aug 9. Calculate its z-score.

2 pts
- 5

Skew, and a check. (a) Use Table 2 to name the **two** variables whose mean is furthest above their median relative to their spread, and say what that tells you about their shape. (b) r between rainfall_mm and pastry_units is **-0.09**. When rainfall_mm is added to Model 4 (Table 4), its t-statistic is **-2.5**; after the **11 days with 5 mm or more** are removed ($n = 157$) it is **-1.7**. What does this tell you about rainfall as a predictor? (c) day_of_week is in column B. Write the Sheets formula for row 2 that derives **weekend** (1 for Saturday or Sunday, 0 otherwise).

3 pts
- 6

An extreme day. On Aug 9 (Saturday, beach_event = 1, temperature_c 21.6, wind 8.3 km/h) pastry sales were **103**. Model 4 predicts **84** for that day and its SE of y is **8.4**. Refitting Model 4 without that day changes the weekend coefficient from 11.30 to 10.91 and R^2 from 0.407 to 0.386. (a) How many SEs is its residual? (b) Delete it, keep it, or something else? Justify.

2 pts

STAGE 3 · Check how the variables relate to each other

≈ 8 min · 7 pts

Table 3. Correlations (Sheets: CORREL) between pastry_units and the candidate predictors, all 168 days.

	units	temp	sun	rain	wind	humid	event	wkend	price
temperature_c	-0.26	1							
sunshine_hours	-0.20	0.78	1						
rainfall_mm	-0.09	-0.15	-0.33	1					
wind_kmh	-0.14	-0.06	-0.01	0.00	1				
humidity_pct	-0.01	-0.46	-0.42	0.47	0.10	1			
beach_event	0.22	0.19	0.09	-0.13	0.16	-0.16	1		
weekend	0.50	-0.02	-0.02	0.10	-0.08	0.05	0.01	1	
pastry_price_eur	-0.13	0.56	0.40	-0.17	0.07	-0.25	0.30	0.00	1
post_change	-0.13	0.56	0.40	-0.17	0.07	-0.25	0.30	0.00	1.00

- 7 Find the patterns.** (a) Which one predictor has the strongest linear relationship with pastry_units? Give its r . (b) List **every pair of predictors** with $|r| > 0.7$ (ignore the sign). 2 pts

- 8 Overlap.** (a) pastry_price_eur and post_change have $r = 1.00$. Why are they so similar? (b) What does this mean for estimating the effect of the 30 June price rise from these data? (c) For the weather pair in 7(b), which variable would you keep, and why? 3 pts

- 9 A 0/1 predictor by hand.** Table 2 gives the mean (0.29) and standard deviation (0.45) of weekend. Calculate the simple-regression slope of pastry_units on weekend ($b = r \times s_y \div s_x$, with r from Table 3) and the intercept ($a = \bar{y} - b \times \bar{x}$). Say in plain words what the slope and the intercept mean. 2 pts

STAGE 4 · Build and compare models with LINEST

≈ 14 min · 13 pts

Table 4. Five models of pastry_units fitted with LINEST(..., TRUE, TRUE) to all 168 days (k = number of predictors).

Model	Predictors	k	R ²	SE of y
M1	temperature_c	1	0.068	10.45
M2	temperature_c, sunshine_hours	2	0.068	10.48
M3	temperature_c, weekend, beach_event	3	0.379	8.58
M4	temperature_c, weekend, beach_event, wind_kmh	4	0.407	8.42
M5	temperature_c, sunshine_hours, rainfall_mm, wind_kmh, humidity_pct, beach_event, weekend, pastry_price_eur	8	0.437	8.30

Table 5. The raw LINEST output for Model 4 (pastry_units on temperature_c, weekend, beach_event, wind_kmh; X columns in that order), all 168 days.

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	-0.26	7.67	11.30	-1.00	88.43
Row 2	0.10	1.62	1.44	0.19	4.75
Row 3	0.41	8.42	#N/A	#N/A	#N/A
Row 4	27.92	163.00	#N/A	#N/A	#N/A
Row 5	7913.43	11549.71	#N/A	#N/A	#N/A

- 10 Decode and be surprised.** (a) Match each number in Row 1 to its variable (or the intercept). (b) Calculate the t-statistic for temperature_c and interpret its coefficient in one sentence. It has the opposite sign to the one most people expect: give one possible real-world reason. (c) How many days were used to fit this model? Show how you know from the output.

3 pts

- 11 A variable you left out.** Your intuition said wind should not matter, but Model 4 adds wind_kmh to Model 3. (a) Calculate the t-statistic for wind_kmh. (b) Calculate the adjusted R² for M3 and M4 (n = 168). (c) With eight candidate predictors, about one test in twenty will look significant by chance. How would you decide whether wind earns a place?

3 pts

- 12 A price effect you cannot isolate.** The café raised the pastry price from €2.50 to €2.75 on 30 June. Daily pastry units: **before** (n = 84) mean 68.5, sd 10.9; **after** (n = 84) mean 65.6, sd 10.5. (a) Calculate the SE of the difference and t; is it significant? (b) Does “not significant” mean the price rise had no effect? (c) In Model 5 the price coefficient is -5.61 with a standard error of 6.48. Why is it so imprecise?

3 pts

- 13 Predict, then check.** Use Model 4 (Table 5) to predict pastry_units for each day below and calculate the residual. Call a day **unusual** if |residual| > 2 × SE of y (8.4). What would you do about an unusual day?

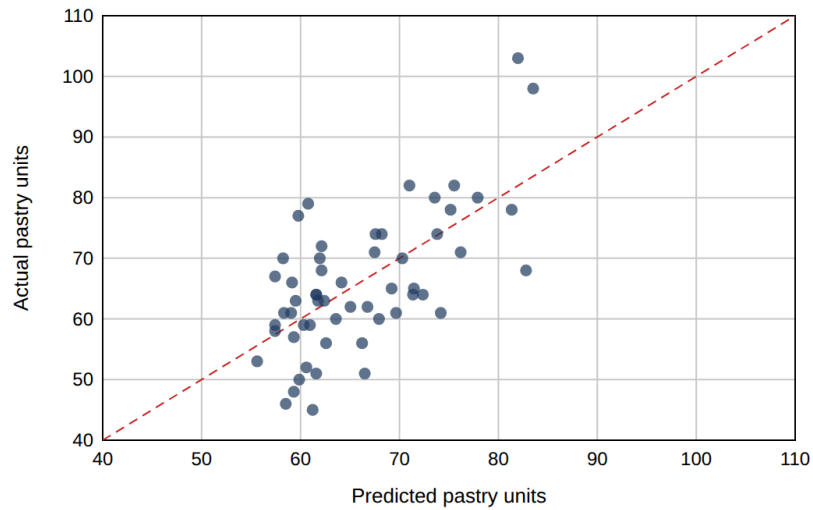
2 pts

Day	Model 4 inputs	Actual	Predicted
Jul 16	temperature_c = 22.9, weekend = 0, beach_event = 0, wind_kmh = 8	61	
Jun 26	temperature_c = 22.9, weekend = 0, beach_event = 0, wind_kmh = 8.9	45	

- 14 Actual versus predicted.** Figure 2 plots the **actual** pastry units against the units **predicted by Model 3** (refitted on the first 112 days) for the 56 later days that the model did not see. The dashed line is perfect prediction. (a) Describe the biggest miss: is the model over- or under-predicting on that day, and by roughly how much? (b) What would it mean if most dots sat below the dashed line?

2 pts

Figure 2. Actual against predicted pastry units, 56 unseen days (Model 3).



STAGE 5 · Does the model forecast unseen days?

≈ 8 min · 7 pts

Models 3 and 5 were **refitted using only the first 112 days**. Their R^2 on those 112 days is **0.344** (M3) and **0.443** (M5). Table 6 shows 4 of the 56 later days. The **baseline** uses no model: a **weighted average of the last three same-weekday sales** with weights **0.5** (one week ago), **0.3** (two weeks ago) and **0.2** (three weeks ago).

Table 6. Four unseen days (pastry units).

Day	Actual	1 week ago	2 weeks ago	3 weeks ago	M3 predicts	M5 predicts
Aug 2	80	61	83	61	73.6	72.0
Aug 11	50	77	59	54	59.9	55.3
Aug 19	61	71	57	51	58.3	52.4
Sep 4	59	70	62	74	60.3	59.1

Table 7. Space for your errors.

- 15 Compute the forecast errors.** (a) Calculate the baseline forecast for each day. (b) Complete Table 7 with the absolute error of each forecaster on each day and the **MAE** of each. (1 mark for the baseline forecasts, 2 for the errors, 2 for the MAEs.)

5 pts

Day	Baseline forecast	error baseline	error M3	error M5
Aug 2				
Aug 11				
Aug 19				
Sep 4				
MAE	—			

- 16 Fit versus forecasting.** M5 has the higher R^2 on the training days. (a) Why does that not guarantee better forecasts of unseen days? (b) Which model would you give the owner, and why? Does the best model clearly beat the baseline?

2 pts

STAGE 6 · Advise the owner and extend the model

≈ 8 min · 7 pts

- 17 Advice for the owner.** Write 3–4 sentences to the owner, who has no statistics background. Say which variables you recommend for forecasting pastry sales, how far off a typical forecast will be (use a number from this practice), and one thing the model cannot tell the owner.

3 pts

- 18 Explain the surprise.** The owner says: “Hot days are good for business, so pastries should sell *more* in summer, but your model says the opposite.” Give **two different explanations** (one about the real world, one about the data, measurement or another variable), say which you would test first and how (LINEST and unseen days), and what result would change your mind.

4 pts

Reference sheet (you may use this on any question)**z-score** = $(\text{value} - \text{mean}) \div s$ · treat $|z| > 2$ as unusual**t-statistic** = coefficient $\div$ its standard error · treat $|t| > 2$ as statistically significant (5% level) **r^2** = share of variation shared by two variables · two predictors with $|r| > 0.7$ overlap heavily**Simple regression from summaries:** slope $b = r \times s_y \div s_x$ · intercept $a = \bar{y} - b \times \bar{x}$ · prediction $\hat{y} = a + b \times x$ **Residual** = actual – predicted · **MAE** (mean absolute error) = average of the absolute residuals**Adjusted R^2** = $1 - (1 - R^2)(n - 1) \div (n - k - 1)$, n = number of days, k = number of predictors**Difference of two means:** $SE = \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$, $t = (\text{mean}_2 - \text{mean}_1) \div SE$ **Weighted average** = $\Sigma(\text{weight} \times \text{value}) \div \Sigma(\text{weights})$ **LINEST(y, X, TRUE, TRUE)** with k predictor columns returns 5 rows: Row 1 coefficients **in reverse order** (last X column first, ... first X column, intercept last) · Row 2 their standard errors · Row 3: R^2 , SE of y (typical size of a residual) · Row 4: F, df (= $n - k - 1$) · Row 5: SS regression, SS residual (#N/A fills the empty cells)**Scratch space**

— End of practice —